Arbiter PUF Based Mutual Authentication and Session Key Agreement Scheme for IoT

Thesis is submitted in the partial fulfillment

of the requirements for the degree of

MASTER OF TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING

By

SAURABH SONI

24CS4131

Under the guidance of

Dr. Subhas Chandra Sahana

Designation(S)

Department of Computer Science Engineering



National Institute of Technology

Durgapur, India

MAY 2026



DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY
DURGAPUR, INDIA

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Durgapur
May, 2026

Saurabh Soni
Roll No.- 24CS4131

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY
DURGAPUR, INDIA

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Professor and Head Department of Computer Science Engineering National Institute of Technology Durgapur	Dr. Subhas Chanda Sahana Professor Department of Computer Science Engineering National Institute of Technology Durgapur

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NATIONAL INSTITUTE OF TECHNOLOGY

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ACKNOWLEDGEMENTS

Foremost, I would like to express my sincere gratitude to my supervisor ***Dr. Subhas Chandra Sahana*** *Professor, Department of Computer Science Engineering, National Institute of Technology Durgapur* for enlightening me the first glance of research, and for his patience, motivation, enthusiasm, and immense knowledge. His inspiring guidance, systematic approach, sensible criticisms and close support throughout the course of this project work helped me in overcoming the problems at many critical stages of the assignment leading and enabling towards a successful accomplishment of this project.

It gives me great pleasure in acknowledging the support and help of Miss. Ananya Banerjee (my PhD senior) to clarify my long technical queries. I am also thankful to my classmates and friends for their love and support.

Finally, I feel great reverence for all my family members and the Almighty, for their blessings and for being a constant source of encouragement.

Durgapur **Saurabh Soni**

May, 2026 Roll No.- 24CS4131

Department of Computer Science Engineering

NIT Durgapur

ABSTRACT

In recent years, the rapid growth of Internet of Things (IoT) devices has introduced significant challenges in ensuring secure and reliable authentication. Traditional security mechanisms rely on storing secret keys in memory, which are vulnerable to physical attacks, cloning, and unauthorized access. To overcome these limitations, hardware-based security techniques such as Physical Unclonable Functions (PUFs) have emerged as a promising solution.

This thesis presents the design and implementation of an Arbiter Physical Unclonable Function (APUF)-based mutual authentication and secure communication system for IoT environments. The proposed system utilizes inherent manufacturing variations in hardware to generate unique and device-specific responses, eliminating the need for permanent key storage. A multi-challenge verification mechanism employing seven challenge–response pairs with threshold-based verification is used to improve authentication robustness and reduce false acceptance and rejection rates.

The system is implemented using FPGA-based APUF modules and a server-side Challenge–Response Pair (CRP) database. During authentication, the server generates different sets of random challenges for two IoT devices. Each device computes corresponding responses using the APUF, and the server verifies the authenticity of both devices by comparing the responses with stored CRPs using majority voting.

After successful mutual authentication, the server securely exchanges response information between the devices to enable independent generation of a common session key. The generated session key is used for AES-based encryption and decryption of transmitted data, ensuring confidentiality and integrity during communication. Timestamp-based validation is also incorporated to prevent replay attacks and ensure session freshness.

The proposed framework was further validated through real-time socket-based communication between two IoT nodes. Experimental results demonstrate high reliability, near-ideal uniformity, strong uniqueness, and efficient hardware resource utilization. The system successfully establishes secure encrypted communication between authenticated devices.

The proposed system provides a lightweight, secure, and scalable authentication framework suitable for resource-constrained IoT applications while addressing key challenges associated with traditional cryptographic approaches.

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SYMBOLS & ABBREVIATIONS

- **Abbreviations**

Abbreviation	Full Form
APUF	Arbiter Physical Unclonable Function
PUF	Physical Unclonable Function
IoT	Internet of Things
CRP	Challenge–Response Pair
FPGA	Field Programmable Gate Array
LUT	Look-Up Table
UART	Universal Asynchronous Receiver Transmitter
AES	Advanced Encryption Standard
SK	Session Key

- **Symbols**

Symbol	Description
C	Challenge input to PUF
R	Response output from PUF
$f(C)$	PUF function mapping challenge to response
q	Number of challenges used in authentication
R_i	Response for ith challenge
HD	Hamming Distance
HW	Hamming Weight
n	Number of bits in response

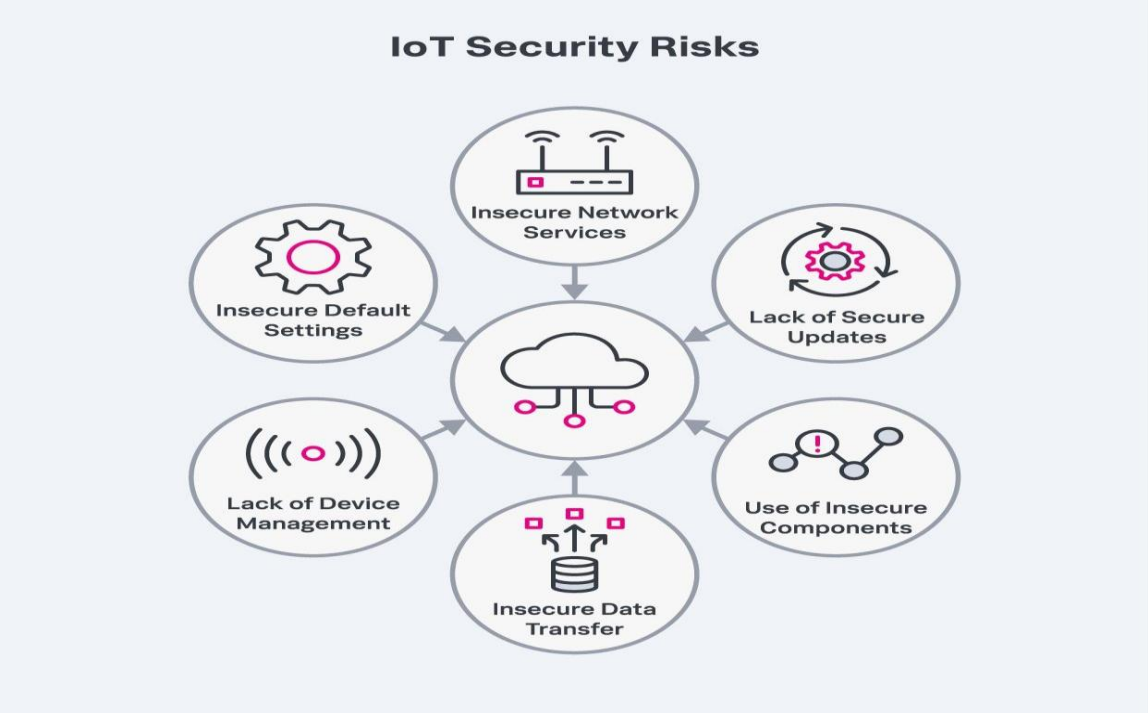
CHAPTER 1

INTRODUCTION

1.1 Background of Hardware Security

With the rapid advancement of embedded systems and the widespread adoption of Internet of Things (IoT) devices, ensuring secure hardware-level authentication has become increasingly important. Modern devices are interconnected and often operate in resource-constrained environments, making them vulnerable to various security threats such as physical tampering, cloning, and unauthorized access.

Traditional security mechanisms rely on storing secret keys in memory (either volatile or non-volatile). However, these stored keys are susceptible to attacks such as side-channel analysis, memory probing, and reverse engineering. Once the secret key is extracted, the entire security system becomes compromised.



1.1 Iot Security Risks

To overcome these challenges, hardware-based security techniques have been introduced, which utilize intrinsic physical characteristics of integrated circuits to provide security. These techniques aim to eliminate the dependency on stored secrets and instead generate secure identifiers directly from the hardware itself.

1.2 Motivation for PUF-Based Authentication

The limitations of conventional cryptographic systems in resource-constrained environments motivate the need for lightweight and secure authentication mechanisms. IoT devices, smart cards, and embedded systems often lack the computational capability and secure storage required for traditional cryptographic approaches.

A major concern in such systems is the secure storage of secret keys. If keys are stored in memory, they can be extracted using invasive or non-invasive attacks. This creates a significant vulnerability in authentication systems.

Physical Unclonable Functions (PUFs) provide an alternative solution by generating device-specific responses based on inherent manufacturing variations. These variations are uncontrollable and unique to each device, making it practically impossible to replicate the exact behavior of a hardware instance.

The motivation behind this work is to design a secure authentication system that:

- Eliminates the need for permanent key storage
- Provides strong resistance against cloning and physical attacks
- Is lightweight and suitable for FPGA-based IoT systems

1.3 Overview of Physical Unclonable Function (PUF)

A Physical Unclonable Function is a hardware security primitive that generates a unique response for a given input challenge based on the inherent physical variations introduced during the manufacturing process.

These microscopic variations in integrated circuits result in unpredictable and device-specific behavior. When a challenge is applied to a PUF, it produces a corresponding response, forming a Challenge–Response Pair (CRP). This behavior can be represented as:

$$R = f(C)$$

where C is the challenge, R is the response, and f represents the unique physical characteristics of the device.

PUFs are widely used for:

- Device authentication
- Secure key generation
- Hardware identification
- Anti-counterfeiting applications

Among various PUF architectures, the Arbiter Physical Unclonable Function is widely used due to its simple structure and efficient implementation on FPGA platforms.

1.4 Problem Statement

Despite the advancements in cryptographic security, existing authentication mechanisms face significant challenges in hardware-constrained environments. The primary issue lies in the secure storage and management of secret keys, which are vulnerable to extraction and cloning attacks.

Additionally, many existing PUF implementations focus only on circuit-level design and performance evaluation, without integrating them into a complete system-level authentication framework. This limits their practical applicability in real-world IoT systems.

Therefore, there is a need to develop a secure and efficient authentication system that:

- Does not rely on stored secret keys
- Provides strong device uniqueness and reliability
- Supports system-level authentication with real-time verification

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1.5 Objectives of the Work

The main objectives of this thesis are as follows:

To design and implement an Arbiter PUF on FPGA

• To generate and analyze Challenge–Response Pairs (CRPs)

• To evaluate PUF performance using metrics such as uniformity, reliability, and uniqueness

• To develop a secure mutual authentication framework based on PUF

• To implement a multi-challenge authentication mechanism for improved robustness

• To establish session key agreement between authenticated IoT devices

• To implement AES-based secure communication using the generated session key

• To analyze hardware resource utilization for lightweight implementation

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1.6 Scope of the Work

This work focuses on the design and implementation of a PUF-based mutual authentication and secure communication system using FPGA technology. The scope includes:

• Implementation of Arbiter PUF using Verilog on FPGA

• Generation and collection of CRPs using UART communication

• Evaluation of PUF performance metrics

• Development of a server-assisted mutual authentication mechanism

• Demonstration of authentication using multi-challenge verification

• Session key generation using CRP-derived responses

• AES-based encrypted communication between IoT devices

• Real-time socket communication for system-level validation

The work is limited to FPGA-based implementation and does not include extensive analysis under varying environmental conditions such as temperature and voltage**.**

1.7 Organization of the Thesis

The remainder of this thesis is organized as follows:

- Chapter 2 presents a detailed literature review on PUF architectures, evaluation metrics, and existing authentication mechanisms.
- Chapter 3 describes the problem formulation and the proposed PUF-based authentication system, including architecture and implementation details.
- Chapter 4 discusses the experimental results, performance evaluation, and system-level validation.
- Chapter 5 concludes the thesis and outlines potential directions for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to PUF-Based Security

With the rapid growth of IoT and embedded systems, ensuring secure and lightweight authentication has become a critical challenge. Conventional cryptographic methods rely on secret key storage, which is vulnerable to physical attacks such as memory extraction and reverse engineering.

To overcome these limitations, hardware-based security mechanisms such as Physical Unclonable Function have been introduced. PUFs exploit inherent manufacturing variations in integrated circuits to generate unique and unpredictable responses for a given input challenge.

As discussed in , PUF-based security provides tamper-resistant authentication without requiring permanent key storage, making it highly suitable for resource-constrained IoT environments. Additionally, PUFs offer resistance against cloning and physical attacks due to their intrinsic randomness.

2.2 Types of PUFs

PUFs can be broadly classified based on their underlying physical principles:

1. Delay-Based PUFs

These PUFs exploit variations in signal propagation delays.
Examples include:

- Arbiter PUF
- Ring Oscillator PUF

Among these, the Arbiter Physical Unclonable Function is widely used due to its simplicity and scalability.

2. Memory-Based PUFs

These rely on the startup behavior of memory elements.
Examples:

- SRAM PUF
- DRAM PUF

3. Other PUF Types

- Optical PUF
- Coating PUF
- Composite PUF

According to , delay-based PUFs are more suitable for FPGA implementations due to their lightweight nature and ease of design.

PUF Type	Principle	Advantage	Limitation
Arbiter PUF	Delay-based	Simple, scalable	ML attack risk
RO PUF	Frequency Variation	High randomness	Power consumption
SRAM PUF	Memory startup	Easy implementation	Limited CRPs

Table 2.1: Comparison of Different Types of Physical Unclonable Functions

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2.3 Arbiter PUF Architecture

The Arbiter PUF is a delay-based architecture that utilizes propagation delay differences between two identical signal paths.

As described in :

- It consists of multiple switch (MUX) stages
- A challenge controls the switching configuration
- Two signals propagate through parallel paths
- An arbiter (D flip-flop) determines the faster signal

The output response depends on which signal arrives first, forming a unique challenge–response relationship.

The mathematical representation is:

$$R = f(C)$$

Arbiter PUFs are widely used due to:

- Simple hardware implementation
- Large number of CRPs (strong PUF)
- Suitability for authentication systems

2.4 PUF Evaluation Metrics

The performance of a PUF is evaluated using standard statistical metrics:

1. Uniformity

Measures the balance between '0's and '1's in responses.
Ideal value ≈ 50%.

2. Reliability

Measures stability of responses under repeated measurements.
Ideal value ≈ 100%.

3. Uniqueness

Measures inter-device variation for the same challenge.
Ideal value ≈ 50%.

Metric	Definition	Ideal Value
Uniformity	Balance of 0's and 1's	~50%
Reliability	Stability of response	~100%
Uniqueness	Difference across devices	~50%

Table 2.2: Standard Performance Metrics for Evaluating PUF Designs

As highlighted in , these metrics are computed using Hamming distance and are essential for validating PUF quality. Environmental factors such as temperature and voltage variations can affect reliability, requiring careful design considerations.

2.5 Existing Authentication Mechanisms Using PUF

Several authentication mechanisms using PUF have been proposed in the literature:

- In , a lightweight authentication protocol for IoT-based wireless sensor networks is proposed using PUF and machine learning models. The system provides two-way authentication and reduces computation overhead.
- In , a PUF-based authentication scheme (PMsec) is developed for Internet of Medical Things (IoMT), which eliminates key storage and enhances system robustness.
- Other approaches use:
    - Symmetric key cryptography with PUF-generated keys
    - Elliptic Curve Cryptography (ECC)-based authentication
    - Blockchain-integrated PUF authentication systems

These methods demonstrate that PUFs can effectively replace traditional key storage mechanisms and enable secure authentication in IoT environments.

Parameter	Details
Software Platform	Xilinx ISE
Programming Language	Verilog
FPGA Device	Artix-7 FPGA
Uniqueness	51.34%
Uniformity	57.64%
Reliability	97.57%
LUTs	389
MUXs	66
Slices	132

Table 2.3: Existing APUF Implementation and Performance Metrics

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2.6 Limitations of Existing Systems

Despite their advantages, existing PUF-based authentication systems have several limitations:

1. CRP Storage Overhead
   Many systems rely on storing large numbers of CRPs on a server, which creates scalability and security issues .

2. Vulnerability to Machine Learning Attacks
   Strong PUFs such as Arbiter PUF can be modeled using machine learning techniques when sufficient CRPs are available.

3. Environmental Sensitivity
   Variations in temperature, voltage, and noise can affect reliability.

4. High Computational Complexity
   Some protocols involve heavy cryptographic operations, making them unsuitable for resource-constrained devices.

5. Lack of System-Level Validation
   Many works focus only on circuit-level design and do not demonstrate complete authentication systems.

2.7 Research Gap

From the literature review, the following research gaps are identified:

- Existing works focus mainly on PUF design or protocol design separately, rather than integrating both into a complete system
- Heavy dependence on CRP storage, leading to scalability issues
- Limited work on multi-challenge authentication mechanisms for improving reliability
- Lack of practical FPGA-based implementation with full system validation

To address these gaps, this thesis proposes:

- An FPGA-based implementation of Arbiter PUF
- A complete authentication framework including CRP enrollment and verification
- A multi-challenge authentication mechanism to improve robustness

CHAPTER 3

PROBLEM FORMULATION AND PROPOSED METHODOLOGY

3.1 Problem Definition

With the rapid growth of IoT and embedded systems, ensuring secure device authentication has become a major challenge. Traditional authentication mechanisms rely on storing secret keys in memory, which are vulnerable to physical attacks such as key extraction, cloning, and tampering. Resource-constrained devices further limit the feasibility of complex cryptographic techniques.

To address these issues, a hardware-based authentication mechanism is required that eliminates the need for permanent key storage while providing strong security. In this work, an authentication system based on Arbiter Physical Unclonable Function is proposed to generate device-specific responses using intrinsic hardware variations.

3.2 Overview of Proposed Authentication System

The proposed system integrates an FPGA-based APUF with a server-side authentication mechanism. Instead of storing secret keys, the system utilizes challenge–response pairs (CRPs) generated from the physical characteristics of the hardware.

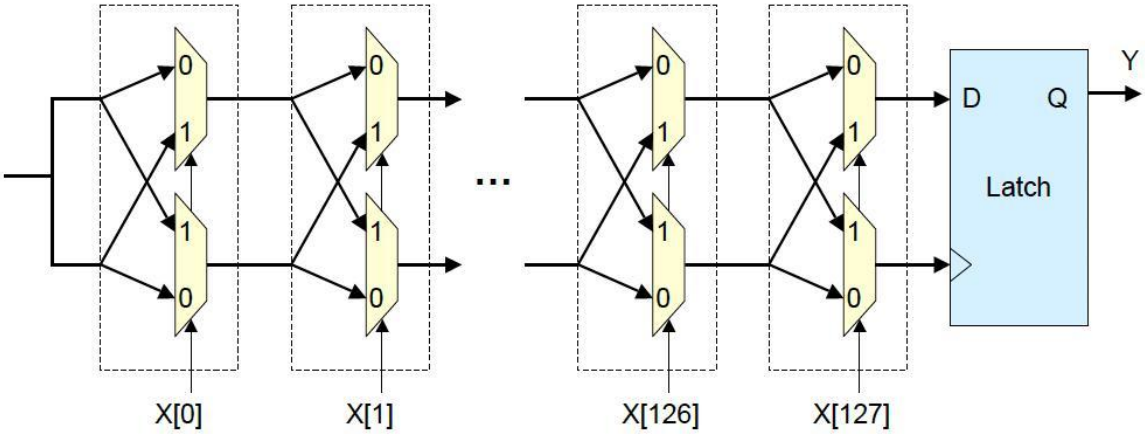
The authentication process consists of two phases:

- Enrollment Phase: CRPs are generated and stored securely in the server database
- Authentication Phase: The device is verified by comparing real-time responses with stored CRPs

To enhance reliability and reduce false rejection, a multi-challenge verification mechanism is employed.

3.3 Architecture of Arbiter PUF

The Arbiter PUF is a delay-based hardware structure that exploits variations in signal propagation delays. It consists of:



3. 1 The structure of an APUF

- A chain of multiplexers controlled by challenge bits
- Two parallel signal paths
- An arbiter (D flip-flop)

A rising signal is applied simultaneously to both paths. Due to manufacturing variations, one signal arrives earlier than the other. The arbiter determines the output based on arrival time:

- Output = 1 → upper path faster
- Output = 0 → lower path faster

This race condition produces a unique response for each device.

3.4 Internal Design of Implemented APUF

The implemented APUF consists of an n-stage switching structure realized on FPGA. Each stage contains multiplexers controlled by individual challenge bits.

For an n-bit challenge vector:

- Each bit determines the switching configuration
- Signals propagate through configurable delay paths
- Final response is generated by the arbiter

Due to routing and manufacturing variations in FPGA fabric, each instance produces a unique mapping between challenges and responses.

3.5 Challenge–Response Pair (CRP) Generation

A CRP is defined as a pair consisting of:

- Challenge (input bit sequence)
- Response (output bit generated by APUF)

Mathematically:

$$R = f(C)$$

Where:

- C= Challenge
- R= Response
- f= Physical function of the device

A large number of CRPs are generated to uniquely identify the device.

3.6 CRP Enrollment Phase

During enrollment, a set of random challenges is applied to the FPGA-based APUF. The corresponding responses are generated and collected.

- Challenges are applied multiple times

- Stable responses are recorded
- CRPs are stored in a secure server database

The IoT device does not store any secret key, improving resistance against memory-based attacks.

3.7 Multi-Challenge Authentication Mechanism

To improve authentication reliability, a multi-challenge verification scheme is used.

Instead of relying on a single CRP:

- A set of $q$challenges is selected
- Responses are generated for each challenge
- Server compares responses with stored CRPs

A match score is computed:

- If score ≥ threshold → ACCEPT
- Otherwise → REJECT

This approach:

- Reduces false rejection
- Improves robustness
- Handles noise and instability

3.8 Server-Side CRP Database Architecture

The system includes a trusted server that maintains a CRP database for each device.

- Stores enrolled CRPs
- Sends challenges during authentication

- Verifies responses

This architecture ensures:

- No key storage in device
- Secure authentication
- Resistance to cloning attacks

3.9 System Workflow

The complete system operates in the following steps:

1. IoT-A sends authentication request to server
2. Server generates different random challenge sets for IoT-A and IoT-B
3. Challenges are applied to FPGA-based APUF modules
4. IoT devices generate challenge–response pairs
5. Responses are transmitted to server
6. Server validates timestamp and verifies responses using stored CRPs
7. Threshold-based majority voting is performed
8. If both devices are authenticated, server exchanges response information
9. Both devices independently generate the same session key
10. AES-based secure communication is established between IoT devices

3.10 Hardware Implementation

The proposed system is implemented using:

- FPGA Board: Basys-3 (Artix-7)
- Design Language: Verilog

- Tool: Xilinx Vivado

Modules implemented:

- Challenge Generator
- MUX Delay Chain
- Arbiter (Flip-Flop)
- UART Interface (for CRP transfer)

The UART module enables communication between FPGA and PC for storing and analyzing CRPs.

3.11 Mutual Authentication and Session Key Agreement Protocol

The proposed system extends conventional PUF-based authentication by implementing mutual authentication between two IoT devices along with secure session key establishment.

The protocol involves three entities:

• IoT-A (Initiating Device)

• IoT-B (Target Device)

• Trusted Authentication Server

The complete authentication process consists of the following message exchange steps:

Step 1: Authentication Request (m1)

IoT-A initiates communication by sending an authentication request to the server along with device identities and timestamp information.

m1: IoT-A → Server

{ID_A, ID_B, Timestamp}

Step 2: Challenge Distribution to IoT-A (m2)

The server generates a random set of seven challenges for IoT-A.

m2: Server → IoT-A

{CA = [C1A, C2A, ..., C7A]}

Step 3: Challenge Distribution to IoT-B (m3)

The server generates a different challenge set for IoT-B.

m3: Server → IoT-B

{CB = [C1B, C2B, ..., C7B]}

Step 4: Response Generation by IoT-A (m4)

IoT-A applies the received challenges to the APUF and generates responses.

m4: IoT-A → Server

{RA = [R1A, R2A, ..., R7A]}

Step 5: Response Generation by IoT-B (m5)

IoT-B similarly generates responses using its challenge set.

m5: IoT-B → Server

{RB = [R1B, R2B, ..., R7B]}

Step 6: Server Verification

The server compares the received responses with enrolled CRPs stored in the database.

A threshold-based majority voting mechanism is used:

• If match count ≥ 5 out of 7 → ACCEPT

• Otherwise → REJECT

Both IoT-A and IoT-B must satisfy the threshold condition for successful mutual authentication.

Step 7: Session Key Assistance (m6 and m7)

After successful authentication:

• Server forwards RB to IoT-A

• Server forwards RA to IoT-B

m6: Server → IoT-A

{RB}

m7: Server → IoT-B

{RA}

Step 8: Session Key Generation

Both devices independently generate the same session key using:

SK = SHA256(RA || RB || Timestamp)

The generated session key is unique for every session and depends on responses from both authenticated devices.

Step 9: Secure Communication

The generated session key is used for AES-based encryption and decryption.

IoT-A encrypts a message and transmits it to IoT-B through socket communication.

IoT-B decrypts the received message using the same session key.

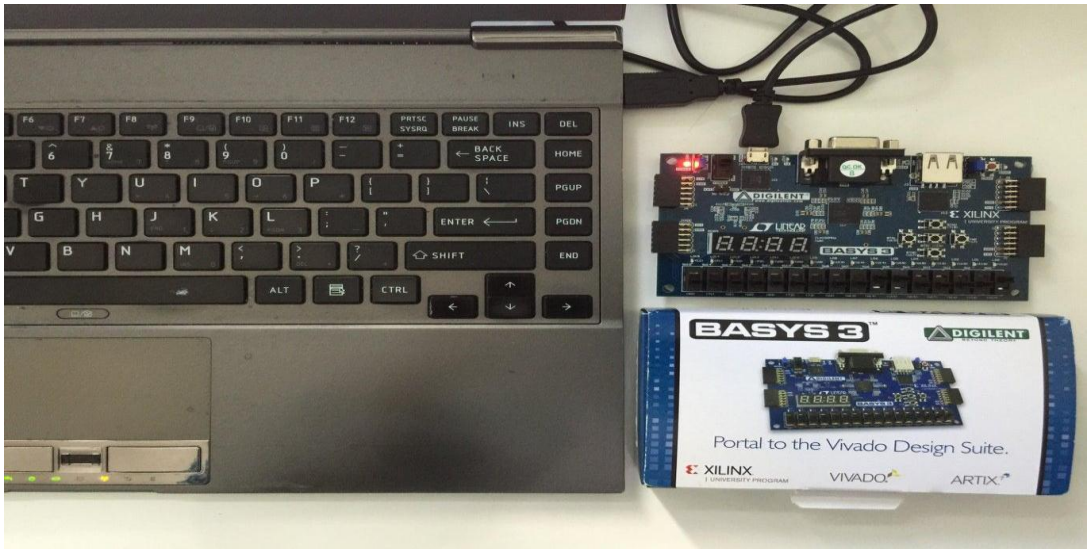
This completes secure communication establishment between authenticated IoT devices.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Experimental Setup

The proposed Arbiter Physical Unclonable Function (APUF)-based authentication system was implemented on an FPGA platform to evaluate its performance and feasibility. The implementation was carried out using the following setup:



4.1 Real setup of Experimental setup

- **Hardware Platform:** Basys-3 FPGA (Artix-7)
- **Design Language:** Verilog
- **Software Tool:** Vivado
- **Communication Interface:** UART for CRP transfer

A set of random binary challenges was applied to the APUF, and the corresponding responses were collected through the UART interface and stored for further analysis.

4.2 CRP Collection and Evaluation Procedure

To evaluate the performance of the APUF, multiple challenge–response pairs (CRPs) were generated.

- A set of *M* random challenges was applied

- Each challenge was repeated *T* times
- Responses were recorded for stability analysis

The collected CRPs were used to compute standard PUF evaluation metrics such as uniformity, reliability, and uniqueness.

4.3 Performance Metrics

The quality of the implemented PUF is evaluated using three standard metrics:

4.3.1 Uniformity

Uniformity measures the balance between '0's and '1's in the response bits.
An ideal value is approximately 50%.

4.3.2 Reliability

Reliability indicates the stability of responses when the same challenge is applied multiple times.
An ideal value is close to 100%.

4.3.3 Uniqueness

Uniqueness measures how different the responses are across multiple PUF instances.
An ideal value is approximately 50%, but higher values indicate strong distinguishability.

4.4 Hardware Resource Utilization

The hardware efficiency of the proposed APUF design was analyzed based on FPGA resource consumption.

Design Type	LUTs	MUXs	Slices	FPGA Platform
Existing APUF Design	389	66	132	Artix-7 FPGA
Proposed Single APUF	**102**	**127**	**26**	**Basys-3 Artix-7**
Proposed Dual APUF	**81**	**--**	**21**	**Basys-3 Artix-7**

Table 4.1: Hardware Resource Utilization of APUF

These results indicate that the design is **lightweight and area-efficient**, making it suitable for low-power and resource-constrained devices such as IoT systems.

4.5 Results of Single APUF

The performance of the single APUF implementation is summarized as follows:

- **Uniformity:** 50.23%
- **Reliability:** 98.98%

The uniformity value is very close to the ideal 50%, indicating a well-balanced distribution of response bits. The high reliability demonstrates that the PUF produces stable responses under repeated evaluations.

4.6 Results of Dual APUF

To enhance security and uniqueness, a Dual APUF architecture was implemented.

- **Uniformity (APUF1):** 51.40%
- **Uniformity (APUF2):** 48.60%
- **Reliability:** 92.61%
- **Uniqueness:** 100%

Design	Uniformity (%)	Uniqueness (%)	Reliability (%)	FPGA Device
APUF	55.69	18.37	99.76	Virtex-5
APUF	42.34	36.75	98.48	Virtex-5
APUF	37.30	8.00	99.70	Virtex-5
APUF	57.64	51.34	97.57	Artix-7
APUF without tuning	43.38	28.00	96.66	Spartan-3
APUF with tuning	48.30	45.25	95.93	Spartan-3
APUF	--	15.15	98.97	Artix-7
APUF	88.90	9.42	99.50	Artix-7
APUF	48.45	44.30	96.00	Spartan-6
FF-APUF	--	41.53	97.10	Artix-7
2-XOR-APUF	5.32	5.39	--	Virtex-5
3-XOR-APUF	54.96	6.35	--	Virtex-5
XOR-APUF	50.73	48.69	99.41	Artix-7
PCS-APUF	57.64	51.34	97.57	Artix-7
Arbiter-Based Multi-PUF	37.03	40.60	--	Artix-7
Multi-line APUF	49.77	46.53	99.88	Virtex-5
DA-APUF	43.38	41.89	97.22	Spartan-6
Proposed Single APUF	**50.23**	--	**98.98**	**Basys-3 Artix-7**

Proposed Dual APUF	**51.40 / 48.60**	**100**	**92.61**	**Basys-3 Artix-7**

Table 4.2: Performance Evaluation of Single and Dual APUF Architectures

The results show that both APUF instances maintain near-ideal uniformity. The uniqueness value of 100% indicates excellent distinguishability between different PUF instances. Although reliability is slightly reduced compared to the single APUF, it remains within acceptable limits.

4.7 Authentication System Evaluation

The proposed system was evaluated in a real-time mutual authentication and secure communication environment.

The implemented framework consists of:

• FPGA-based APUF devices

• Server-side CRP database

• Socket-based communication framework

• AES-based secure communication module

The authentication process follows a multi-stage protocol involving challenge distribution, response generation, threshold-based verification, session key establishment, and encrypted communication.

During authentication:

• IoT-A sends an authentication request to the server

• The server generates different challenge sets for IoT-A and IoT-B

• Both devices generate responses using their FPGA-based APUFs

• The server verifies responses using enrolled CRPs

• Timestamp validation is performed to prevent replay attacks

• Upon successful verification, both devices independently derive the same session key

• AES-based encrypted communication is established

Experimental results demonstrated:

• Successful mutual authentication of legitimate devices

• Successful rejection of invalid responses

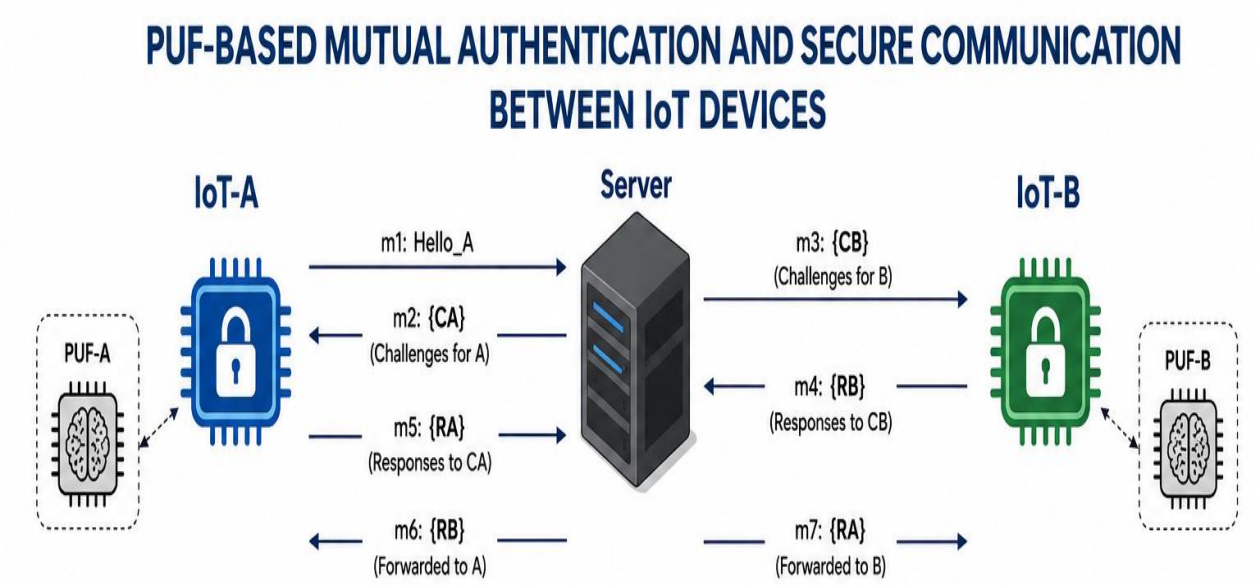
• Correct generation of identical session keys at both devices

• Successful encryption and decryption of transmitted messages

• Stable real-time communication using socket programming

The implementation validates that the proposed system successfully integrates FPGA-based PUF authentication with secure communication mechanisms.

4.8 System-Level Authentication Demonstration:



4.2 Auth system working

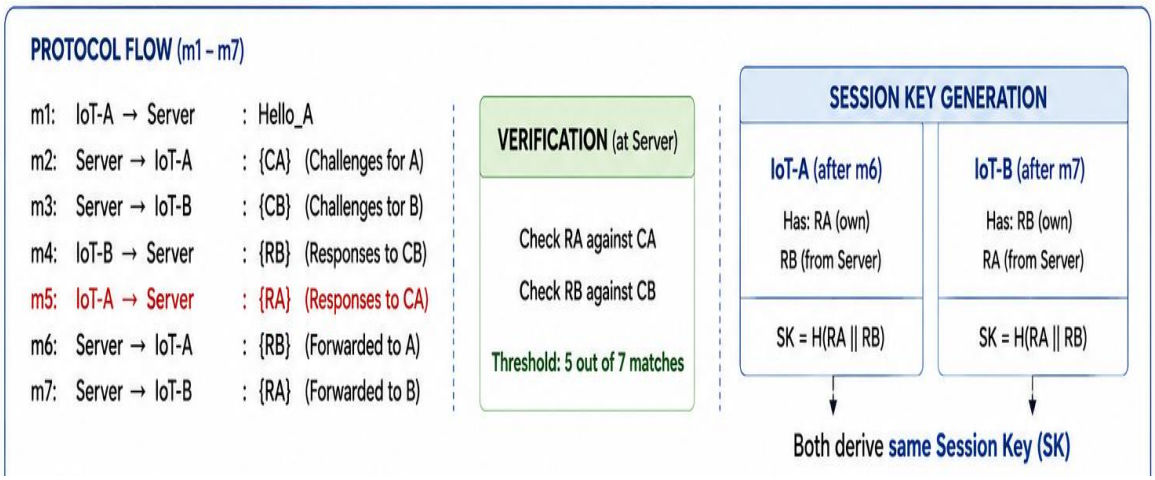
A complete IoT-based authentication framework was implemented, consisting of:

- FPGA-based APUF device
- Server with CRP database
- Communication interface

During authentication:

- The server sends challenges
- The device computes responses
- The server verifies and grants access

This validates that the proposed system is not limited to circuit-level design but also works effectively at the system level.



4.3 Working Explanation of Auth System

4.9 Discussion of Results

The experimental results demonstrate that the implemented APUF-based authentication system achieves a good balance between security, reliability, and hardware efficiency.

- Uniformity values are close to ideal → good randomness
- High reliability ensures stable operation
- 100% uniqueness ensures strong device identification
- Low resource utilization confirms suitability for embedded systems

However, minor variations in reliability indicate sensitivity to internal FPGA routing variations, which can be further analyzed in future work.

The proposed authentication framework provides multiple security advantages over conventional key-storage-based systems.

1. Resistance to Cloning Attacks

The use of FPGA-based APUF responses generated from intrinsic hardware variations makes device cloning extremely difficult.

2. Resistance to Key Extraction Attacks

The proposed system does not store permanent secret keys inside IoT devices. Session keys are dynamically generated after authentication.

3. Replay Attack Protection

Timestamp validation is incorporated during authentication. Authentication requests outside the predefined time window are rejected.

4. Mutual Authentication

Both IoT devices are independently verified by the server using different challenge sets and threshold-based response matching.

5. Secure Communication

AES-based encryption

CHAPTER 5

CONCLUSION AND SCOPE OF FUTURE WORK

5.1 Conclusion

In this thesis, a hardware-based authentication system using Arbiter Physical Unclonable Function has been successfully designed and implemented on an FPGA platform. The proposed system leverages intrinsic manufacturing variations to generate unique and device-specific responses, eliminating the need for storing secret keys in memory.

The implemented APUF architecture was evaluated using standard performance metrics such as uniformity, reliability, and uniqueness. The obtained results demonstrate that the system achieves near-ideal uniformity and high reliability, indicating stable and balanced response generation. Furthermore, the Dual APUF implementation enhances uniqueness, ensuring strong distinguishability between devices.

In addition to circuit-level design, a complete authentication framework was developed by integrating CRP enrollment, server-side verification, and a multi-challenge authentication mechanism. This approach significantly improves authentication robustness and reduces the probability of false acceptance or rejection.

The hardware implementation on FPGA shows low resource utilization, confirming that the proposed design is lightweight and suitable for resource-constrained environments such as IoT devices. Overall, the proposed system demonstrates that PUF-based authentication provides a secure, efficient, and scalable alternative to traditional key-based security mechanisms.

5.2 Key Contributions

The major contributions of this work are summarized as follows:

- Design and implementation of an FPGA-based APUF architecture
- Generation and analysis of Challenge–Response Pairs (CRPs)
- Evaluation using standard PUF metrics (uniformity, reliability, uniqueness)

- Implementation of Dual APUF for improved uniqueness
- Development of a multi-challenge authentication mechanism
- Integration of a complete system-level authentication framework with server-side verification
- Demonstration of lightweight and efficient hardware implementation

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5.3 Limitations

Despite the promising results, the proposed system has certain limitations:

- Performance under environmental variations (temperature, voltage, aging) was not extensively analyzed
- Arbiter PUFs are susceptible to machine learning-based modeling attacks when a large number of CRPs are exposed
- Slight variations in reliability may occur due to internal FPGA routing differences

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5.4 Future Scope

The proposed work can be further extended in several directions:

- Environmental Analysis: Evaluate PUF performance under varying temperature and voltage conditions
- Attack Resistance: Improve resistance against machine learning attacks using advanced PUF designs or obfuscation techniques
- Secure Communication: Integrate encryption protocols with PUF-based authentication for end-to-end secure communication
- Strong PUF Design: Explore Feed-Forward or XOR Arbiter PUF architectures for enhanced non-linearity
- Real-World Deployment: Implement the system in large-scale IoT networks for practical validation

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APPENDIX

Appendix A: Experimental Setup

- FPGA Board: Basys-3 (Artix-7)
- Software Tool: Xilinx Vivado
- Programming Language: Verilog
- Communication Interface: UART

Appendix B: Sample Challenge–Response Data
Example of generated CRPs used for evaluation:

10101010	1
11001100	0
11110000	1
00001111	0

PUBLICATIONS

Lightweight Authenticated Key Agreement Protocol using ECC and PUF for Telecare Medical Information System. (**Communicated**)